Bayesian Optimization Report: Function 1

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.759	Good
2	0.749	Good
3	0.751	Good
4	0.720	Good
5	0.718	Good
6	0.715	Good
7	0.706	Good
8	0.708	Good
9	0.709	Good
10	0.710	Good
11	0.712	Good
12	0.712	Good
13	0.713	Good
14	0.715	Good
15	0.712	Good
16	0.714	Good
17	0.715	Good
18	0.717	Good
19	0.718	Good
20	0.720	Good
21	0.721	Good
22	0.723	Good
23	0.724	Good
24	0.725	Good
25	0.727	Good
26	0.728	Good
27	0.730	Good
28	0.732	Good
29	0.733	Good
30	0.735	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.0005	0.0001	0.0008
2	0.0004	0.0001	0.0008
3	0.0004	0.0000	0.0008
4	0.0004	0.0001	0.0008
5	0.0004	0.0001	0.0008
6	0.0004	0.0000	0.0008
7	0.0004	0.0000	0.0008
8	0.0004	0.0000	0.0007
9	0.0003	0.0000	0.0007
10	0.0003	0.0000	0.0007
11	0.0003	0.0000	0.0007
12	0.0003	0.0000	0.0007
13	0.0003	0.0000	0.0007
14	0.0003	0.0000	0.0007
15	0.0003	0.0000	0.0007
16	0.0003	0.0000	0.0007
17	0.0003	0.0000	0.0007
18	0.0003	0.0000	0.0007
19	0.0003	0.0000	0.0007
20	0.0003	0.0000	0.0007
21	0.0003	0.0000	0.0007
22	0.0003	0.0000	0.0007
23	0.0003	0.0000	0.0007
24	0.0002	0.0000	0.0006
25	0.0002	0.0000	0.0006
26	0.0002	0.0000	0.0006
27	0.0002	0.0000	0.0006
28	0.0002	0.0000	0.0006
29	0.0002	0.0000	0.0006
30	0.0002	0.0000	0.0005

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.73918939 0.6822252]	-0.0035541695528320833	Matern(ν=0.5)	EI	0.759	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(-0.8636279076410789)}}
2	[0.95062971 0.67968147]	-0.0034986434763171394	Matern(ν=0.5)	EI	0.749	None
3	[0.73467275 0.68209884]	-0.00355933823 9725964	Matern(ν=0.5)	EI	0.751	None
4	[0.67750106 0.68102371]	-0.00359730878 3605173	Matern(ν=0.5)	EI	0.720	None
5	[1. 0.67252352]	-0.0034150258979697384	Matern(ν=0.5)	EI	0.718	None
6	[0.80193331 0.67802656]	-0.00356181563 1696134	Matern(ν=0.5)	EI	0.715	None
7	[0.8832558 0.6801096]	-0.00352551169 7884695	Matern(ν=0.5)	EI	0.706	None
8	[0.84211783 0.68040197]	-0.0035370300863560985	Matern(ν=0.5)	EI	0.708	None
9	[0.55111521 0.67479405]	-0.0035350909803058073	Matern(ν=0.5)	EI	0.709	None
10	[0.7015364 0.67869271]	-0.00360253417 2524637	Matern(ν=0.5)	EI	0.710	None
11	[0.80507854 0.67873937]	-0.00356100456 1126102	Matern(ν=0.5)	EI	0.712	None
12	[0.62208848 0.67850122]	-0.00361212103 5802918	Matern(ν=0.5)	EI	0.712	None
13	[1. 0.67535974]	-0.0034868831568759463	Matern(ν=0.5)	EI	0.713	None
14	[1. 0.67715442]	-0.00351071304 229889	Matern(ν=0.5)	EI	0.715	None
15	[0.98318852 0.67963551]	-0.0035111076893417697	Matern(ν=0.5)	EI	0.712	None
16	[0.99016259 0.67825539]	-0.00351478358 64487538	Matern(ν=0.5)	EI	0.714	None
17	[0.66444872 0.67792992]	-0.00361916946 3656096	Matern(ν=0.5)	EI	0.715	None
18	[0.95263424 0.67889319]	-0.00350393856 2311646	Matern(ν=0.5)	EI	0.717	None
19	[0.8219218 0.67812177]	-0.0035530472915072365	Matern(ν=0.5)	EI	0.718	None

20	[0.90267919 0.6781442]	-0.0035236645836843493	Matern(ν=0.5)	EI	0.720	None
21	[0.69996596 0.67639981]	-0.00358982697 3419981	Matern(ν=0.5)	EI	0.721	None
22	[0.59945778 0.68073851]	-0.00358691278 4777555	Matern(ν=0.5)	EI	0.723	None
23	[0.67233503 0.98036565]	2.02738642064125e-05	Matern(ν=0.5)	EI	0.724	None
24	[0.55352118 0.87555278]	1.252399111627625e-05	Matern(ν=0.5)	EI	0.725	None
25	[0.52431112 0.67355262]	-0.0034917844415318484	Matern(ν=0.5)	EI	0.727	None
26	[0.89830316 0.67860884]	-0.00352604747 0838821	Matern(ν=0.5)	EI	0.728	None
27	[0.55423343 0.97782769]	2.843736374539066e-05	Matern(ν=0.5)	EI	0.730	None
28	[0.65736984 0.98505262]	2.477572880547874e-05	Matern(ν=0.5)	EI	0.732	None
29	[0.49794882 0.67794991]	-0.0035480719704923756	Matern(ν=0.5)	EI	0.733	None
30	[0.40647555 0.94585533]	6.652293429019584e-06	Matern(ν=0.5)	EI	0.735	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.001780
EI	0.000000
PI	0.000000
THOMPSON	0.001338
KG	0.000000
ENTROPY	0.697430
PORTFOLIO	0.002393

Iteration 2

Method	Max Value
UCB	0.000521
EI	0.000000
PI	0.000000

THOMPSON	0.000800
KG	0.000000
ENTROPY	0.688671
PORTFOLIO	0.002127

Iteration 3

Method	Max Value
UCB	0.000496
EI	0.000000
PI	0.000000
THOMPSON	0.001375
KG	0.000000
ENTROPY	0.690019
PORTFOLIO	0.001979

Iteration 4

Method	Max Value
UCB	0.000458
EI	0.000000
PI	0.000000
THOMPSON	0.000938
KG	0.000000
ENTROPY	0.661267
PORTFOLIO	0.001676

Iteration 5

Method	Max Value
UCB	0.000457
EI	0.000000
PI	0.000000
THOMPSON	0.001214
KG	0.000000
ENTROPY	0.659866
PORTFOLIO	0.001637

Iteration 6

Method	Max Value
UCB	0.000467
EI	0.000000
PI	0.000000
THOMPSON	0.001222
KG	0.000000
ENTROPY	0.657267
PORTFOLIO	0.001559

Iteration 7

Method	Max Value
UCB	0.000455
EI	0.000000
PI	0.000000
THOMPSON	0.001345
KG	0.000000
ENTROPY	0.648614
PORTFOLIO	0.001497

Iteration 8

Method	Max Value
UCB	0.000442
EI	0.000000
PI	0.000000
THOMPSON	0.001107
KG	0.000000
ENTROPY	0.650178
PORTFOLIO	0.001375

Iteration 9

Method	Max Value
UCB	0.000432
EI	0.000000
PI	0.000000

THOMPSON	0.000774
KG	0.000000
ENTROPY	0.651776
PORTFOLIO	0.001316

Iteration 10

Method	Max Value
UCB	0.000426
EI	0.000000
PI	0.000000
THOMPSON	0.000935
KG	0.000000
ENTROPY	0.652751
PORTFOLIO	0.001220

Iteration 11

Method	Max Value
UCB	0.000413
EI	0.000000
PI	0.000000
THOMPSON	0.001363
KG	0.000000
ENTROPY	0.654095
PORTFOLIO	0.001022

Iteration 12

Method	Max Value
UCB	0.000410
EI	0.000000
PI	0.000000
THOMPSON	0.000766
KG	0.000000
ENTROPY	0.653961
PORTFOLIO	0.001265

Iteration 13

Method	Max Value
UCB	0.000404
EI	0.000000
PI	0.000000
THOMPSON	0.000690
KG	0.000000
ENTROPY	0.655322
PORTFOLIO	0.001130

Iteration 14

Method	Max Value
UCB	0.000403
EI	0.000000
PI	0.000000
THOMPSON	0.001098
KG	0.000000
ENTROPY	0.656620
PORTFOLIO	0.000942

Iteration 15

Method	Max Value
UCB	0.000396
EI	0.000000
PI	0.000000
THOMPSON	0.000997
KG	0.000000
ENTROPY	0.654313
PORTFOLIO	0.000928

Iteration 16

Method	Max Value
UCB	0.000386
EI	0.000000
PI	0.000000

THOMPSON	0.000769
KG	0.000000
ENTROPY	0.655675
PORTFOLIO	0.000898

Iteration 17

Method	Max Value
UCB	0.000393
EI	0.000000
PI	0.000000
THOMPSON	0.000877
KG	0.000000
ENTROPY	0.657256
PORTFOLIO	0.000824

Iteration 18

Method	Max Value
UCB	0.000383
EI	0.000000
PI	0.000000
THOMPSON	0.000662
KG	0.000000
ENTROPY	0.658667
PORTFOLIO	0.000698

Iteration 19

Method	Max Value
UCB	0.000384
EI	0.000000
PI	0.000000
THOMPSON	0.000782
KG	0.000000
ENTROPY	0.660192
PORTFOLIO	0.000615

Iteration 20

Method	Max Value
UCB	0.000375
EI	0.000000
PI	0.000000
THOMPSON	0.000728
KG	0.000000
ENTROPY	0.661560
PORTFOLIO	0.000880

Iteration 21

Method	Max Value
UCB	0.000379
EI	0.000000
PI	0.000000
THOMPSON	0.000648
KG	0.000000
ENTROPY	0.662796
PORTFOLIO	0.000765

Iteration 22

Method	Max Value
UCB	0.000379
EI	0.000000
PI	0.000000
THOMPSON	0.000744
KG	0.000000
ENTROPY	0.664250
PORTFOLIO	0.000730

Iteration 23

Method	Max Value
UCB	0.000372
EI	0.000000
PI	0.000000

THOMPSON	0.001250
KG	0.000000
ENTROPY	0.665441
PORTFOLIO	0.000647

Iteration 24

Method	Max Value
UCB	0.000373
EI	0.000000
PI	0.000000
THOMPSON	0.000774
KG	0.000000
ENTROPY	0.666491
PORTFOLIO	0.000546

Iteration 25

Method	Max Value
UCB	0.000364
EI	0.000000
PI	0.000000
THOMPSON	0.000535
KG	0.000000
ENTROPY	0.668219
PORTFOLIO	0.000536

Iteration 26

Method	Max Value
UCB	0.000349
EI	0.000000
PI	0.000000
THOMPSON	0.000475
KG	0.000000
ENTROPY	0.669315
PORTFOLIO	0.000397

Iteration 27

Method	Max Value
UCB	0.000363
EI	0.000000
PI	0.000000
THOMPSON	0.000517
KG	0.000000
ENTROPY	0.670794
PORTFOLIO	0.000313

Iteration 28

Method	Max Value
UCB	0.000356
EI	0.000000
PI	0.000000
THOMPSON	0.000441
KG	0.000000
ENTROPY	0.672211
PORTFOLIO	0.000373

Iteration 29

Method	Max Value
UCB	0.000357
EI	0.000000
PI	0.000000
THOMPSON	0.000543
KG	0.000000
ENTROPY	0.673916
PORTFOLIO	0.000464

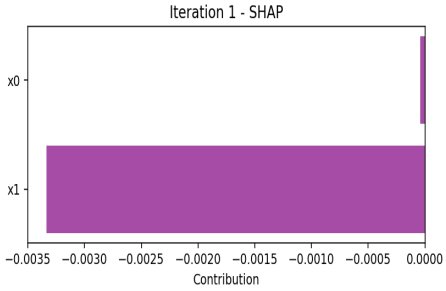
Iteration 30

Method	Max Value
UCB	0.000360
EI	0.000000
PI	0.000000

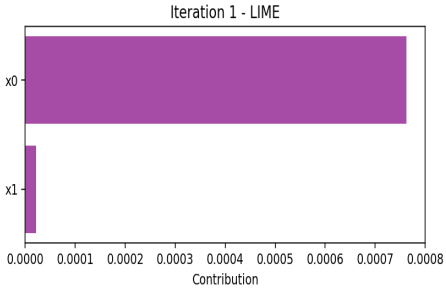
THOMPSON	0.000596
KG	0.000000
ENTROPY	0.675027
PORTFOLIO	0.000495

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



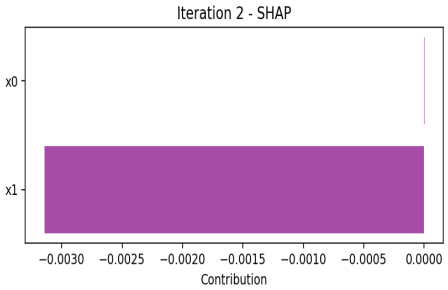
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	-0.00004
x1	-0.00333

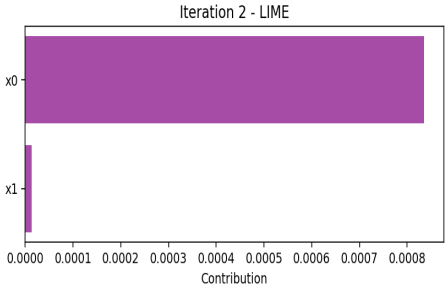
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.00076
x1	0.00002

Iteration 2 - SHAP



Iteration 2 - LIME



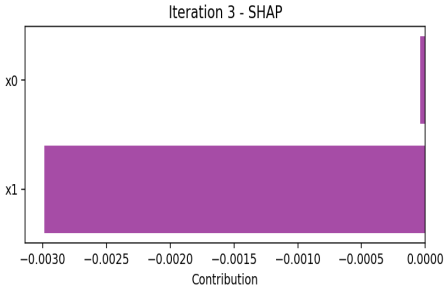
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00314

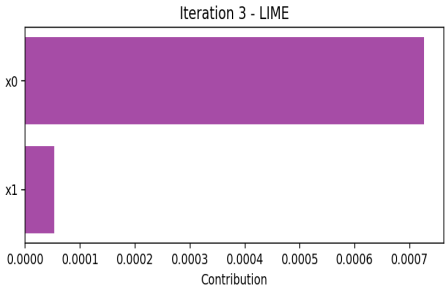
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.00083
x1	0.00001

Iteration 3 - SHAP



Iteration 3 - LIME



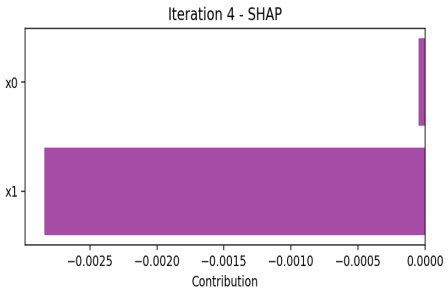
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	-0.00004
x1	-0.00299

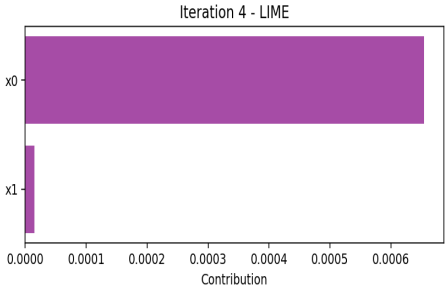
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.00073
x1	0.00005

Iteration 4 - SHAP



Iteration 4 - LIME



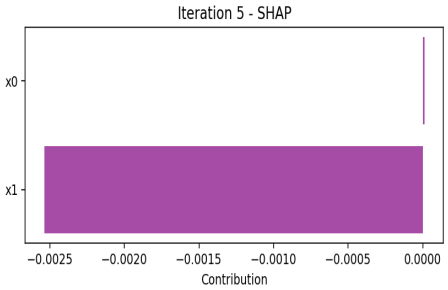
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-0.00005
x1	-0.00284

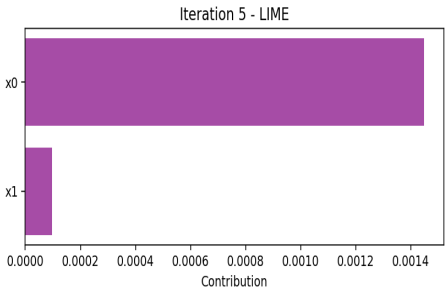
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.00065
x1	0.00001

Iteration 5 - SHAP



Iteration 5 - LIME



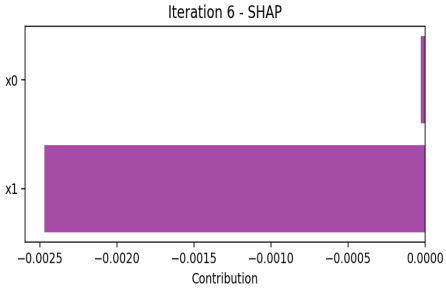
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	0.00001
x1	-0.00254

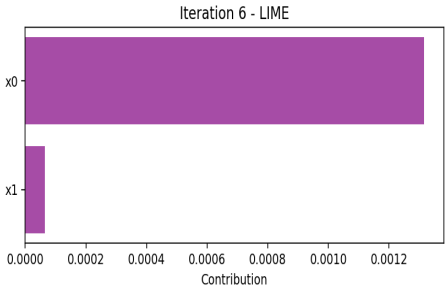
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.00145
x1	0.00010

Iteration 6 - SHAP



Iteration 6 - LIME



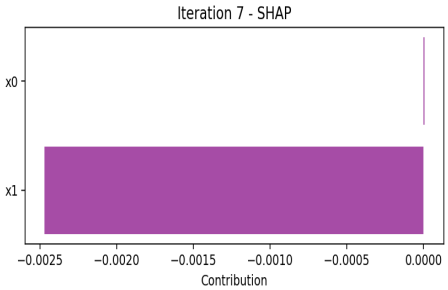
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	-0.00003
x1	-0.00247

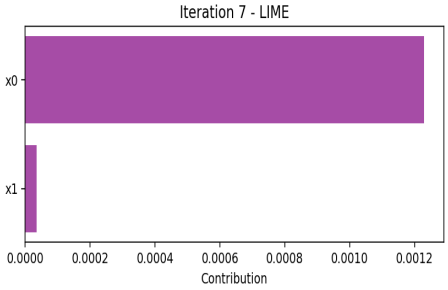
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.00132
x1	0.00006

Iteration 7 - SHAP



Iteration 7 - LIME



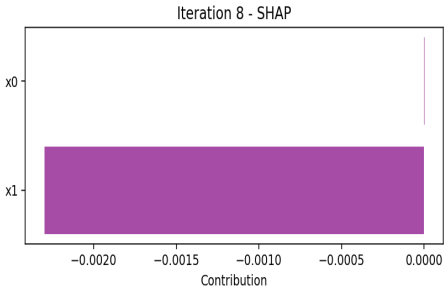
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	0.00001
x1	-0.00247

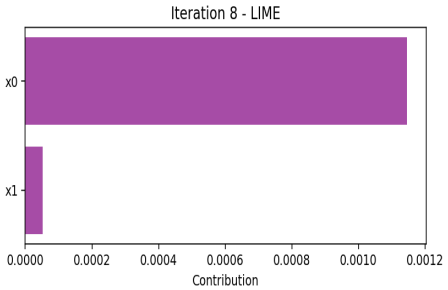
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.00123
x1	0.00003

Iteration 8 - SHAP



Iteration 8 - LIME



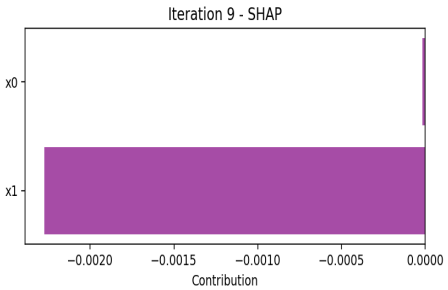
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00230

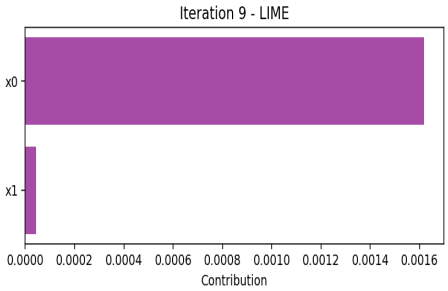
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.00115
x1	0.00005

Iteration 9 - SHAP



Iteration 9 - LIME



Iteration 9 - SHAP Values

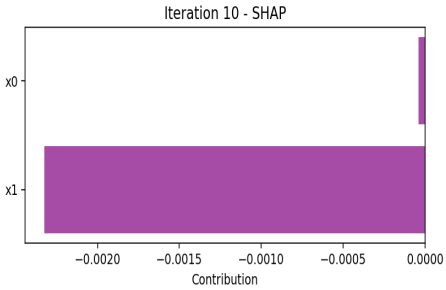
Feature	SHAP Value
x0	-0.00002
x1	-0.00227

Iteration 9 - LIME Values

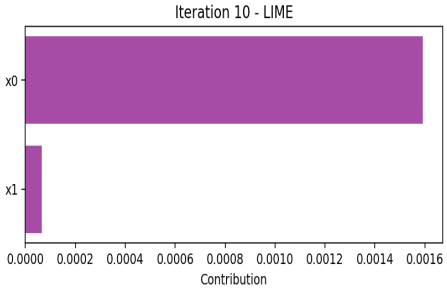
Feature	LIME Value
x0	0.00162

x1	0.00004

Iteration 10 - SHAP



Iteration 10 - LIME



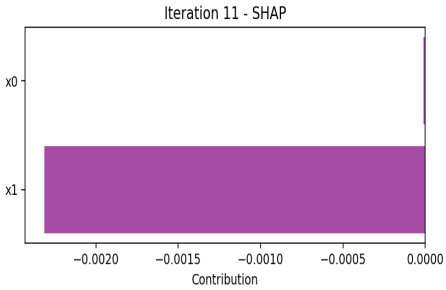
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	-0.00004
x1	-0.00232

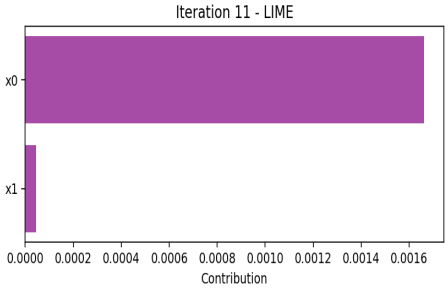
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.00159
x1	0.00007

Iteration 11 - SHAP



Iteration 11 - LIME



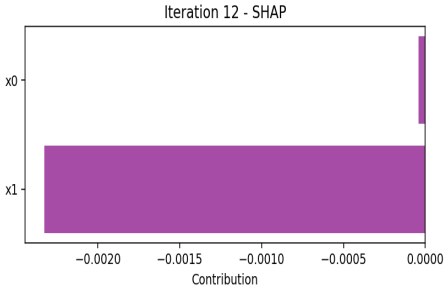
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	-0.00001
x1	-0.00231

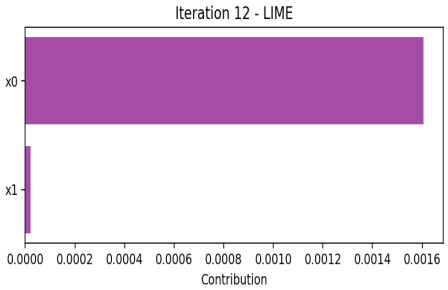
Iteration 11 - LIME Values

Feature	LIME Value
x0	0.00166
x1	0.00004

Iteration 12 - SHAP



Iteration 12 - LIME



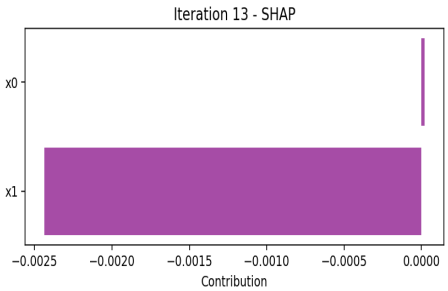
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	-0.00004
x1	-0.00233

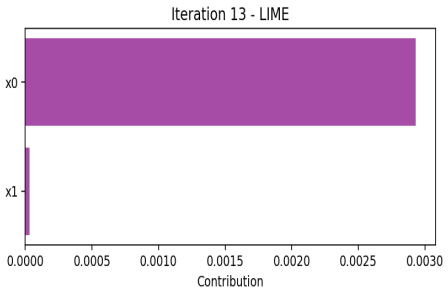
Iteration 12 - LIME Values

Feature	LIME Value
x0	0.00161
x1	0.00002

Iteration 13 - SHAP



Iteration 13 - LIME



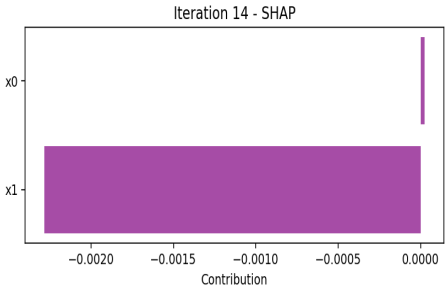
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	0.00002
x1	-0.00244

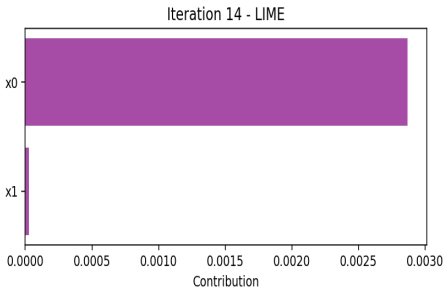
Iteration 13 - LIME Values

Feature	LIME Value
x0	0.00293
x1	0.00003

Iteration 14 - SHAP



Iteration 14 - LIME



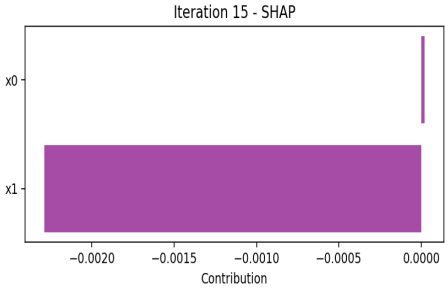
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.00002
x1	-0.00228

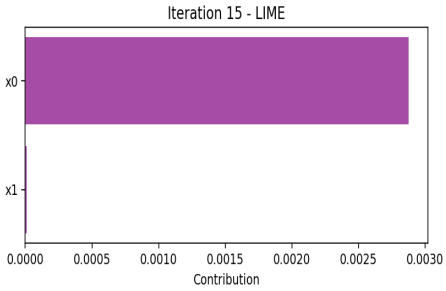
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.00287
x1	0.00003

Iteration 15 - SHAP



Iteration 15 - LIME



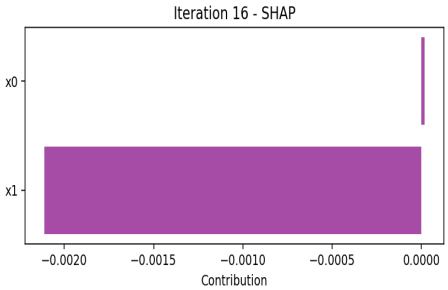
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	0.00002
x1	-0.00229

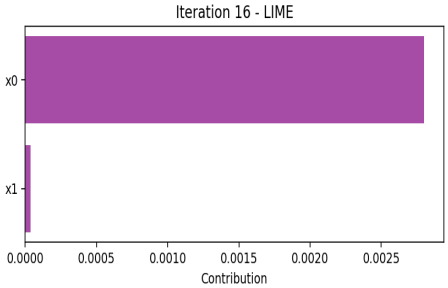
Iteration 15 - LIME Values

Feature	LIME Value
x0	0.00288
x1	0.00001

Iteration 16 - SHAP



Iteration 16 - LIME



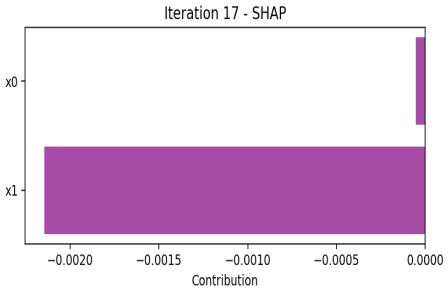
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	0.00002
x1	-0.00211

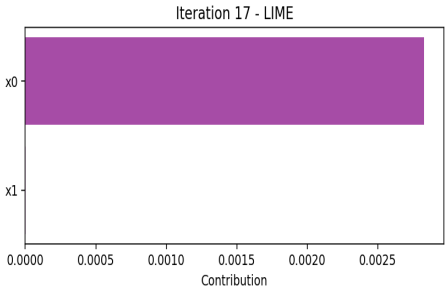
Iteration 16 - LIME Values

Feature	LIME Value
x0	0.00280
x1	0.00004

Iteration 17 - SHAP



Iteration 17 - LIME



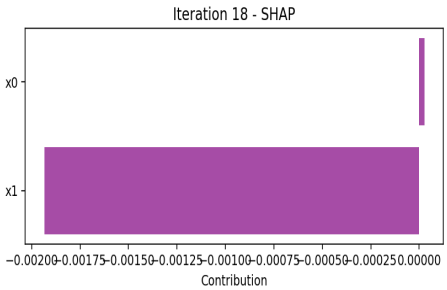
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	-0.00005
x1	-0.00215

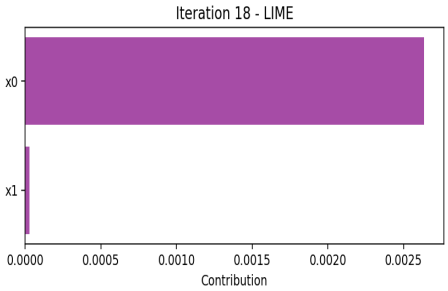
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.00283
x1	0.00000

Iteration 18 - SHAP



Iteration 18 - LIME



Iteration 18 - SHAP Values

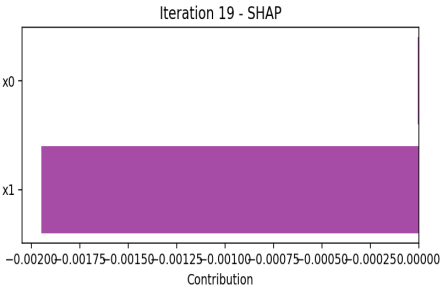
Feature	SHAP Value
x0	0.00003
x1	-0.00193

Iteration 18 - LIME Values

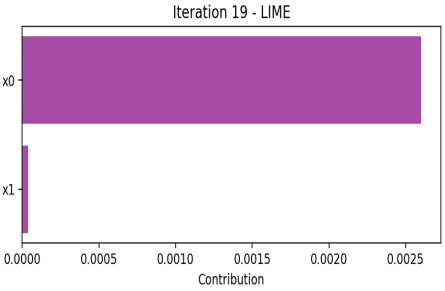
Feature	LIME Value
x0	0.00263

x1	0.00003

Iteration 19 - SHAP



Iteration 19 - LIME



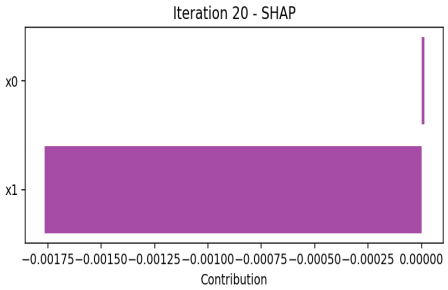
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	-0.00001
x1	-0.00195

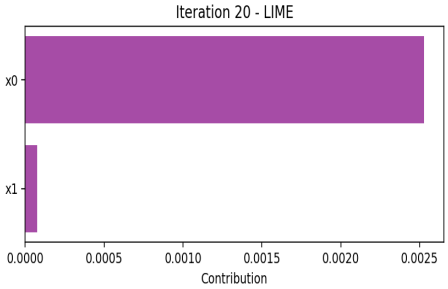
Iteration 19 - LIME Values

Feature	LIME Value
x0	0.00260
x1	0.00004

Iteration 20 - SHAP



Iteration 20 - LIME



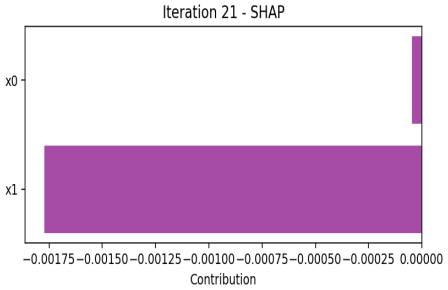
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	0.00001
x1	-0.00177

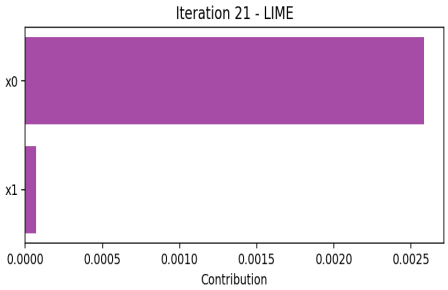
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.00253
x1	0.00007

Iteration 21 - SHAP



Iteration 21 - LIME



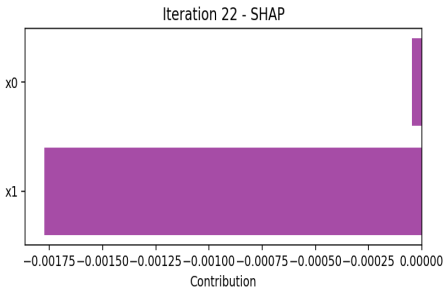
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	-0.00005
x1	-0.00177

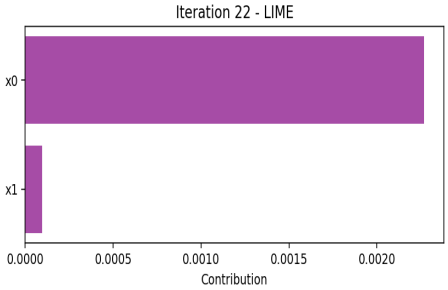
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.00258
x1	0.00007

Iteration 22 - SHAP



Iteration 22 - LIME



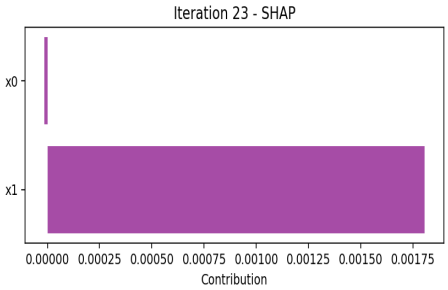
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	-0.00005
x1	-0.00177

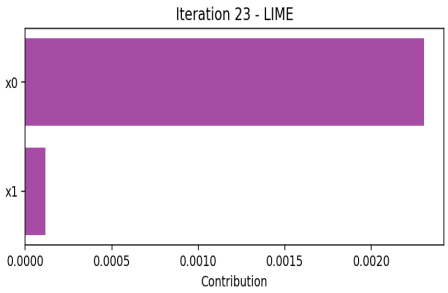
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.00227
x1	0.00010

Iteration 23 - SHAP



Iteration 23 - LIME



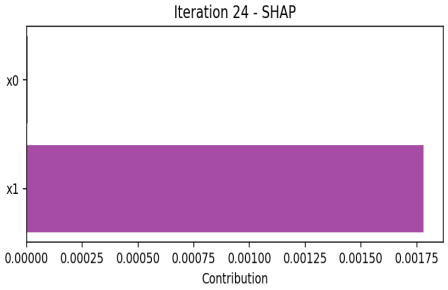
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	-0.00001
x1	0.00181

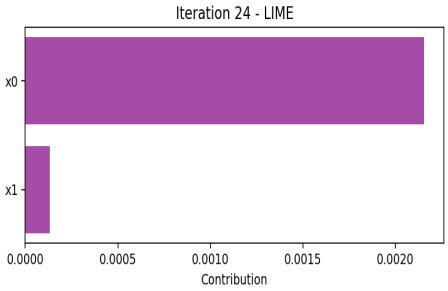
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.00230
x1	0.00012

Iteration 24 - SHAP



Iteration 24 - LIME



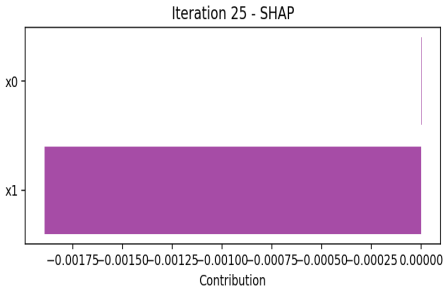
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	0.00001
x1	0.00178

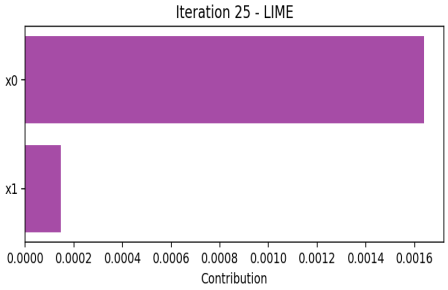
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.00215
x1	0.00013

Iteration 25 - SHAP



Iteration 25 - LIME



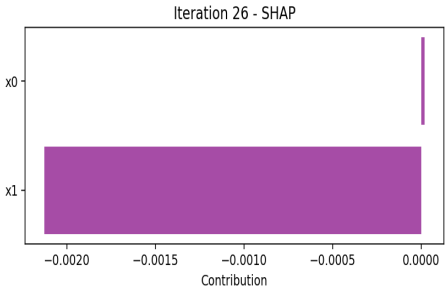
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00189

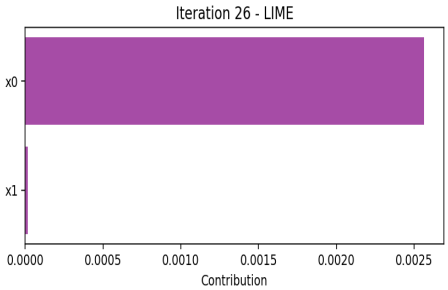
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.00164
x1	0.00015

Iteration 26 - SHAP



Iteration 26 - LIME



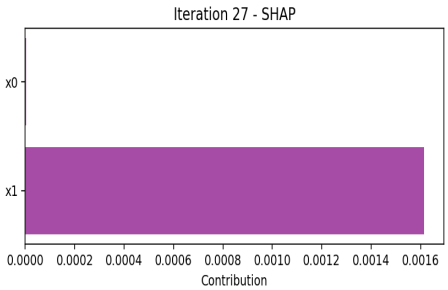
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	0.00002
x1	-0.00213

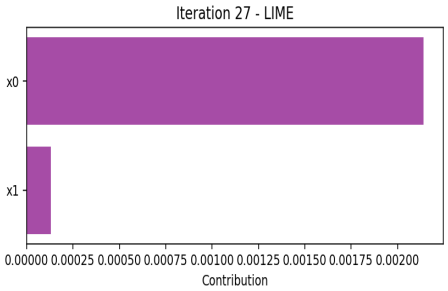
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.00256
x1	0.00002

Iteration 27 - SHAP



Iteration 27 - LIME



Iteration 27 - SHAP Values

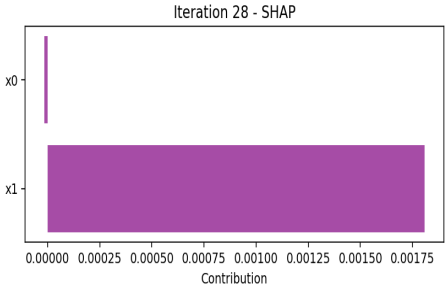
Feature	SHAP Value
x0	0.00000
x1	0.00161

Iteration 27 - LIME Values

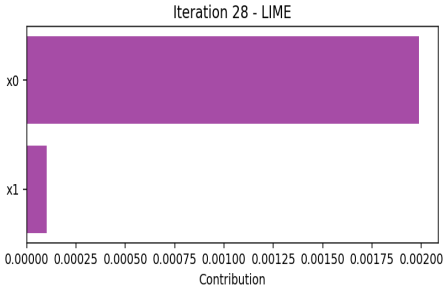
Feature	LIME Value
x0	0.00214

x1	0.00013

Iteration 28 - SHAP



Iteration 28 - LIME



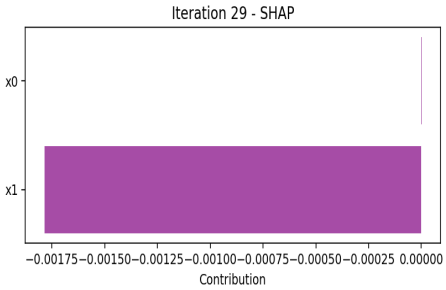
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	-0.00002
x1	0.00181

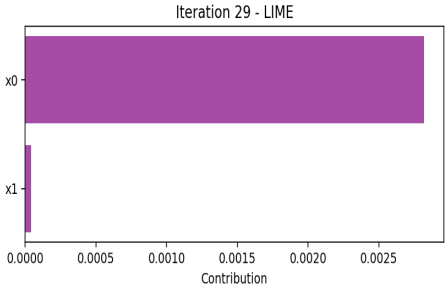
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.00199
x1	0.00010

Iteration 29 - SHAP



Iteration 29 - LIME



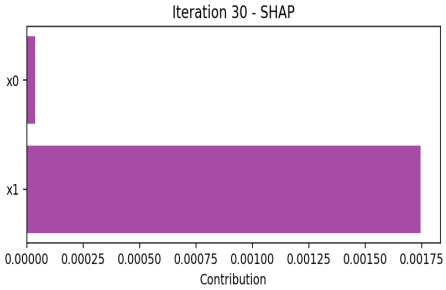
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00179

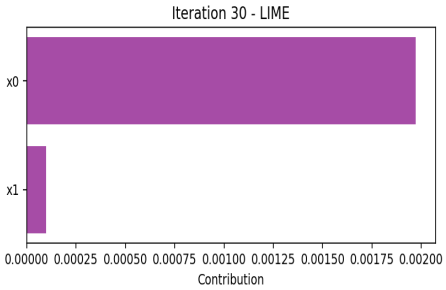
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.00282
x1	0.00004

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	0.00004
x1	0.00175

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.00197
x1	0.00010